tec-speed 30.0 VT-6880

UL Approval: E214381 Version:20/12/2024

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Precautions in Handling

Storage Condition & Shelf Life

		Laminate
Storage Condition	Temperature	Room
	Relative Humidity	/
Shelf Life		24 Months(airproof)

- The grain direction is indicated on the label with an arrow.
- Take care in handling thin core laminates as they are easily damaged.
- Keep work surfaces clean, dry, and completely free of debris.
- During processing, only pick panels by two edges.
- Laminates exceeding 24 months can still be used as long as the copper foil has no severe oxidation or stain.

Designing and Inner layer Process

- Tooling & pinning holes should be drilled or routed. Punching is forbidden.
- Lay the panel on a flat clean surface. The solvent clean should be followed with a clean water rinse.
- Remove copper oxides using commercially available cleaners and micro etchants.
- Abrasive cleaning (e.g., pumice powder or conveyorized brush scrubbing) is not recommended.
- Mechanical surface preparation tends to cause microscopic scratches in copper. These scratches become stress risers that can lead to thermal stress cracks in some environmental conditions. While strongly recommending against abrasive cleaning, we recognize that in some circumstances that abrasive cleaning is necessary. When abrasive cleaning is required, it should be performed with minimal pressure and knowledge of the risks involved. Brush scrubbing may cause material distortion.
- After oxide, it is preferred to press cores within 12 hours. Any cores staged more than 24 hours should be re-treated.

Note: Before inner layer process, it is better to bake at 160 ° C for 4hours.

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Laminates Availability

Laminate	Core Thickness	Size	Dk @ 10GHz	Df @ 10GHz
VT-6880	0.010"(0.254mm)	18”×12”(457mm×305mm) 18”×24”(457mm×610mm)	2.2±0.02	0.0009
	0.020"(0.508mm)		2.2±0.02	0.0009
	0.030"(0.762mm)		2.2±0.02	0.0009

Remarks:

- Copper foil: Hoz~2oz, HTE,RTF and HVLP are all available based upon request.
- More types may be available upon request.
- Metal base and Resistive Cu are available upon request.

Drilling parameters

Drill Bit Diameter		Spindle (KRPM)	Infeed (IPM)	Retract (IPM)	Hit Counts
0.50mm	0.0197in	40	78	500	300
0.65mm	0.0256in	30	60	500	300
1.00mm	0.0394in	20	40	500	300
1.25mm	0.0492in	20	40	500	300
1.59mm	0.0625in	20	40	500	200
2.00mm	0.0797in	20	40	500	200
3.00mm	0.1181in	20	40	500	200

- Carbide drills should be used to minimize wear of cutting edges.
- Chip load: 0.0015”~0.003” / revolution Surface speed: 150~300 SFM.
- Excessive wear of carbide drill bits. Carbide or Diamond coated drill bits are recommended.
- LE Aluminum sheet and white phenolic entry board are preferred, please consult your local Ventec representative for options.

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Desmear Process

Plasma treatment is necessary process, and material should be baked more than 1 hour at 120°C before plasma treatment. Meanwhile, material cannot be suffered any high pressure washing and scrubbing before metallization.

Plasma: Half time of FR-4 conditions is recommended.

Pressure (mT)	Temperature (°C)	Gas mixture	Power (W)	Duration (min)
150	90	80%H_2 , 20% N_2	1800	15
150	95	10%CF_4 , 90% N_2	2000	15
170	95	100% O_2	1800	10
180	95	100% N_2	1800	10

Permanganate Desmear: Not recommended.

Plating and Solder mask

- Conventional electroplating processes can be used. Panels should be handled carefully during racking and unracking to avoid damage or distortion.
- When copper surface preparation is required, chemical cleaning processes are preferred. Mechanical scrubbing should be avoided due to possible mechanical damage. Otherwise, standard processing should be used.
- No special treatment is required if the surface has not been mechanically scrubbed. In rare cases, where adhesion is poor, a plasma treatment may be used to activate the exposed PTFE.

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Routing

Drill Bit Diameter (mm)	Spindle (KRPM)	Infeed (IPM)	Life (Feet)
1.0	20	75	45
2.0	15	65	35
3.0	10	45	30

- Excessive wear of carbide drill bits and Diamond coated drill bits are recommended.
- Rigid phenolic entry and a rigid backer should be used. In some cases, adding paper (white paper or Kraft paper) between the phenolic and the part allows better conformance to surface topography and may reduce burring.
- Materials will generally provide a better edge quality due to none glass fabric.

Packaging and baking recommendation

- It is recommended to bake the board before packaging at 125°C/4~8h to avoid moisture causing a decrease in heat resistance.
- If the PCBs needs to be stored for a long time before use, it is recommended to use aluminum foil vacuum packaging.
- If exceed 3 months after packaging, It is best to bake the PCBs at 125°C/4~6h before assembly before use.